

Chapter 7

Bounded Input Bounded Output Stability

Stability is a deep concept and is broken-down into several sections.

For a stable system, a bounded input (e.g. step and impulse) produces a bounded output. Therefore, this is considered bounded-input bounded-output stability or BIBO stability. A signal is bounded when there is a finite value that the signal magnitude never exceeds.

BIBO stability is analyzed with the open-loop representation of a transfer function by determining the roots of the characteristic polynomial (poles).

When the real value of a pole is negative, then the pole is stable. When all poles of a characteristic polynomial are stable, then the system that produces the signal is stable. If a characteristic polynomial has even one unstable pole then the system is unstable.

Let's look at a few examples.

Example — Stable System

Given:

$$\frac{Y(s)}{R(s)} = \frac{s+4}{s^2+3s+2}$$

Find: The BIBO system stability

A **time domain** approach:

Apply a bounded input, $r(t) = \delta(t)$ and determine the output signal using partial fraction expansion

$$Y(s) = \frac{s+4}{s^2+3s+2} = \frac{s+4}{(s+1)(s+2)} = \frac{k_1}{s+1} + \frac{k_2}{s+2}$$

where k_1 and k_2 are residues — non-zero real numbers in this example.

Then,

$$y(t) = k_1 e^{-t} + k_2 e^{-2t}$$

therefore

$$y(\infty) = k_1 e^{-\infty} + k_2 e^{-\infty} = 0$$

is a bounded output.

Let's choose a different bounded input, set $r(t) = 1$. Then

$$Y(s) = \frac{s+4}{s^2+3s+2} \left(\frac{1}{s} \right) = \frac{s+4}{s(s+1)(s+2)}$$

and as a summation of partials

$$Y(s) = \frac{s+4}{s(s+1)(s+2)} = \frac{k_1}{s} + \frac{k_2}{s+1} + \frac{k_3}{s+2}$$

which produces

$$y(t) = k_1 + k_2 e^{-t} + k_3 e^{-2t}$$

where k_1 , k_2 , and k_3 are residues — non-zero real numbers in this example. Therefore

$$y(\infty) = k_1 + k_2 e^{-\infty} + k_3 e^{-\infty} = k_1$$

which is non-zero, but bounded output.

Example — Unstable System

Given:

$$\frac{Y(s)}{R(s)} = \frac{s+4}{s^2+3s-2}$$

Find: The BIBO system stability

Apply a bounded input, $r(t) = \delta(t)$ and determine the output signal using partial fraction expansion

$$Y(s) = \frac{s+4}{s^2+3s-2} = \frac{s+4}{(s+1)(s-2)} = \frac{k_1}{s+1} + \frac{k_2}{s-2}$$

where k_1 and k_2 are residues — non-zero real numbers in this example. Then,

$$y(t) = k_1 e^{-t} + k_2 e^{2t}$$

which leads to

$$y(\infty) = k_1 e^{-\infty} + k_2 e^{\infty} \rightarrow \infty$$

Now, the first expression $k_1 e^{-\infty} \rightarrow 0$, is a function of a stable pole and by itself is bounded; however, the second expression, $k_2 e^{\infty} \rightarrow \infty$, is a function of an unstable pole and is unbounded. The end result is an unbounded output signal $y(t)$.

Note, in all three examples above there is no need to go into the time domain to determine BIBO stability, one must just observe the poles of the system.

From the first example,

$$\frac{Y(s)}{R(s)} = \frac{B(s)}{A(s)} = \frac{s+4}{s^2+3s+2}$$

Set the characteristic polynomial to zero and solve for the roots

$$s^2 + 3s + 2 = (s + 1)(s + 2) = 0$$

The poles, $s = -1, -2$ are both stable; therefore, the system is BIBO stable.

From the second example,

$$\frac{Y(s)}{R(s)} = \frac{B(s)}{A(s)} = \frac{s + 4}{s^2 + 3s - 2}$$

Set the characteristic polynomial to zero and solve for the roots

$$s^2 + 3s - 2 = 0$$

The second pole, $s = -1, 2$ is unstable; therefore, the system is BIBO is unstable.

When considering an open-loop transfer function

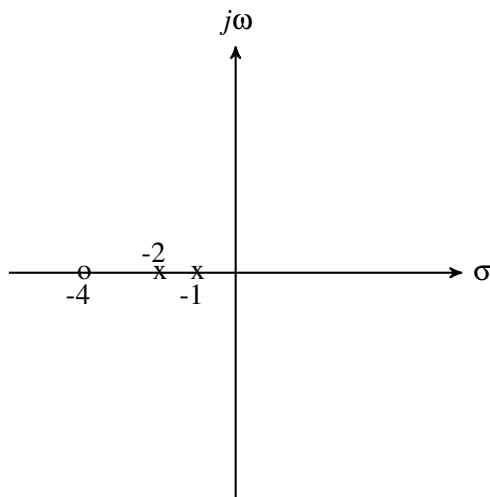
$$\frac{Y(s)}{R(s)} = \frac{B(s)}{A(s)}$$

The roots (zeros) for $B(s)$ shape the dynamic system response to an input, but do not affect stability. The roots (poles) for $A(s)$ shape the dynamic system response and determine the stability of the system. Typically, one plots the zeros and poles in the s-plane. In the plot, zeros are always identified with an “o” symbol and poles with an “x” symbol.

From the stable example above

$$\frac{Y(s)}{R(s)} = \frac{s + 4}{(s + 1)(s + 2)} = \frac{s + 4}{s^2 + 3s + 2}$$

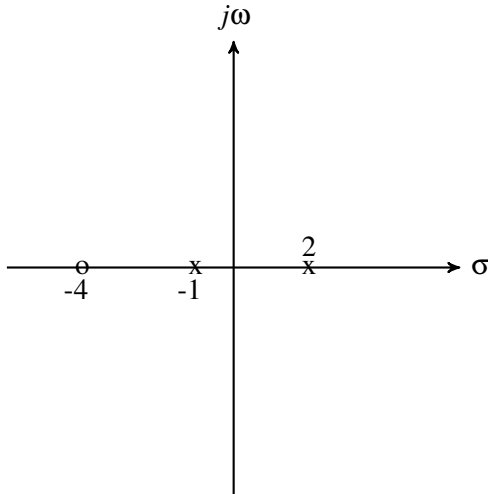
we have an s-plane plot of the zeros and poles



Likewise, from the unstable example above

$$\frac{Y(s)}{R(s)} = \frac{s + 4}{(s + 1)(s - 2)} = \frac{s + 4}{s^2 + 3s - 2}$$

we have an s-plane plot of the zeros and poles



Example — Stable System with Complex Poles

Given:

$$\frac{Y(s)}{R(s)} = \frac{s+4}{s^2+6s+10}$$

Find: The BIBO system stability From the **time domain**, set $r(t) = \delta(t)$

$$Y(s) = \frac{s+4}{s^2+6s+10} = \frac{k_1}{s+3+j} + \frac{k_2}{s+3-j}$$

$$k_1 = \frac{(s+3+j)(s+4)}{(s+3+j)(s+3-j)}, s = -3-j$$

$$k_1 = \frac{(s+4)}{(s+3-j)}, s = -3-j$$

$$k_1 = \frac{(-3-j+4)}{(-3-j+3-j)} = \frac{1-j}{-2j} = \frac{(1-j)2j}{(-2j)2j}$$

$$k_1 = \frac{2+2j}{4} = \frac{1+j}{2}$$

While

$$k_2 = \frac{(s+4)}{(s+3+j)}, s = -3+j$$

$$k_2 = \frac{(-3+j+4)}{(-3+j+3+j)} = \frac{1+j}{2j} = \frac{(1+j)(-2j)}{(2j)(-2j)}$$

$$k_2 = \frac{(2-2j)}{4} = \frac{1-j}{2}$$

Note, $k_1 = k_2^*$. This is always the case for complex numbers, their residues are always complex conjugate pairs.

$$Y(s) = \frac{s+4}{s^2+6s+10} = \frac{k_1}{s+3+j} + \frac{k_1^*}{s+3-j}$$

where the inverse Laplace

$$\mathcal{L}^{-}\{Y(s)\} = k_1 \mathcal{L}^{-}\left\{\frac{1}{s+3+j}\right\} + k_1^* \mathcal{L}^{-}\left\{\frac{1}{s+3-j}\right\}$$

Consider the expressions independently

$$k_1 \mathcal{L}^{-}\left\{\frac{1}{s+3+j}\right\} = k_1 (e^{(-3-j)t}) = k_1 (e^{(-3t)} e^{(-tj)})$$

and

$$k_1^* \mathcal{L}^{-}\left\{\frac{1}{s+3-j}\right\} = k_1^* (e^{(-3+j)t}) = k_1^* (e^{(-3t)} e^{(jt)})$$

where

$$e^{-jt} = \cos(-t) + j \sin(-t) = \cos(t) - j \sin(t)$$

$$e^{jt} = \cos(t) + j \sin(t) = \cos(t) + j \sin(t)$$

The inverse Laplace then simplifies to

$$y(t) = k_1 e^{-3t} (\cos(t) - j \sin(t)) + k_1^* e^{-3t} (\cos(t) + j \sin(t))$$

factor the common exponential function and distribute the residues

$$y(t) = e^{-3t} (k_1 \cos(t) - k_1 j \sin(t) + k_1^* \cos(t) + k_1^* j \sin(t))$$

now, collect like terms

$$y(t) = e^{-3t} ((k_1 + k_1^*) \cos(t)) + (k_1^* - k_1) j \sin(t)$$

which simplifies to

$$y(t) = e^{-3t} \left((k_1 + k_1^*) \cos(t) \right) = e^{-3t} \left(\left(\frac{1+j}{2} + \frac{1-j}{2} \right) \cos(t) \right) = e^{-3t} \cos(t)$$

7.1 Marginal Stability

Stable and unstable poles are straight forward and easily explained using the PFE model and the time domain. Marginally stable poles are *somewhat* more difficult to explain and grasp — but only by a little.

We start with a transfer function with stable poles and a pair of complex conjugate roots represented as a PFE

$$\frac{Y(s)}{R(s)} = \frac{k_1}{s+p_1} + \dots + \frac{k_n}{s+p_n} + \frac{b}{s^2+b^2}$$

The binomial expressions are from the stable poles, the last expression is from the complex roots on the imaginary axis. Set $r(t) = 1$ and solve for $y(t)$

$$Y(s) = \frac{k_0}{s} + \frac{k_1}{s+p_1} + \dots + \frac{k_n}{s+p_n} + \frac{b}{s^2+b^2}$$

Taking the inverse Laplace transform

$$y(t) = k_0 + k_1 e^{-p_1 t} + \dots + k_n e^{-p_n t} + \sin(bt)$$

As $t \rightarrow \infty$

$$y(\infty) = k_0 + \sin(bt)$$

because all of the expressions from the stable poles go to zero as $t \rightarrow \infty$. We can expand this concept to as many complex roots on the imaginary axis as we want. Each pair will generate an additional oscillation expression in the final output, but the output will still be stable.

$$Y(s) = \frac{k_0}{s} + \frac{k_1}{s+p_1} + \dots + \frac{k_n}{s+p_n} + \frac{b_1}{s^2+b_1^2} + \frac{b_2}{s^2+b_2^2}$$

Taking the inverse Laplace transform

$$y(t) = k_0 + k_1 e^{-p_1 t} + \dots + k_n e^{-p_n t} + \sin(b_1 t) + \sin(b_2 t)$$

As $t \rightarrow \infty$

$$y(\infty) = k_0 + \sin(b_1 t) + \sin(b_2 t)$$

Now, we consider a stable system with a single pole at the origin

$$\frac{Y(s)}{R(s)} = \frac{k_1}{s} + \frac{k_2}{s+p_2} + \dots + \frac{k_n}{s+p_n}$$

Let's set the input to an impulse $R(s) = 1$

$$Y(s) = \frac{k_1}{s} + \frac{k_2}{s+p_2} + \dots + \frac{k_n}{s+p_n}$$

The time domain equation

$$y(t) = k_1 + k_2 e^{-p_2 t} + \dots + k_n e^{-p_n t}$$

As $t \rightarrow \infty$

$$y(\infty) = k_1$$

which is a BIBO stable response.

Repeat for a setpoint input

$$Y(s) = \frac{k_1}{s} + \frac{k_2}{s+p_2} + \dots + \frac{k_n}{s+p_n} + \frac{k_m}{s^2}$$

The time domain equation

$$y(t) = k_1 + k_2 e^{-p_2 t} + \dots + k_n e^{-p_n t} + k_m t$$

As $t \rightarrow \infty$

$$\begin{aligned} y(\infty) &= k_1 + k_m(\infty) \\ &= \infty \end{aligned}$$

which is unbounded.

BIBO stable systems with marginally poles:

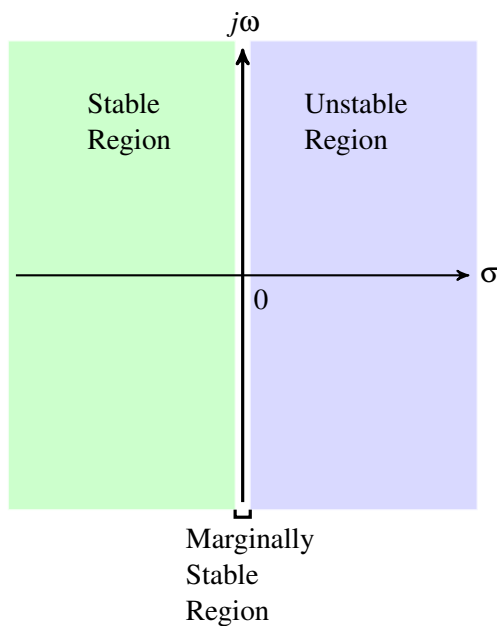
1. Are stable if there is no root at the origin. Complex roots on the $j\omega$ axis produce a oscillating bounded output
2. Are dependent on the input for stability if there is a root at the origin. An impulsive input produces a bounded output; while a setpoint input produces an unstable output
3. Are unstable if there are repeated roots at the origin; regardless of the input

7.2 BIBO Stability Summary

Stability analysis decomposes the s-plane in three regions of stability, as determined by the real axis. A pole is defined as $s = \sigma + j\omega$, where σ is the real part and $j\omega$ is the imaginary part. The three regions of stability are defined as

1. Stable region, $\sigma < 0$
2. Unstable region, $\sigma > 0$
3. Marginally stable region, $\sigma = 0$

The complex component produces the oscillations as observed about as the cosine operator in the previous example, however, the exponential function from the real part drives the solution to zero as $t \rightarrow \infty$.



7.3 Routh-Hurwitz Stability Criterion

Allows us to evaluate the BIBO stability of a system without directly calculating the poles of the system. The process creates an array (more like a table, really) that is then analyzed to determine the BIBO stability based off of the first column. The array is colloquially called the “Routh Array”.

When $A(s)$ is of degree n , the Routh Array will have $n + 1$ rows. The elements of the first two rows have elements composed of the coefficients in $A(s)$.

If $A(s) = a_n s^n + a_{n-1} s^{n-1} + \dots + a_0$ then the first two rows of the array are

Row Label	Col 1	Col 2	Col 3	...
s^n	a_n	a_{n-2}	a_{n-4}	...
s^{n-1}	a_1	a_{n-3}	a_{n-5}	...

the remaining rows ($s^{n-2}, \dots, s^0$) must be calculated.

A complete Routh Array takes on the form

Row Label	Col 1	Col 2	Col 3	...
s^n	a_n	a_{n-2}	a_{n-4}	...
s^{n-1}	a_1	a_{n-3}	a_{n-5}	...
s^{n-2}	x_{31}	x_{32}	x_{33}	...
s^{n-3}	x_{41}	x_{42}	x_{43}	...
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
s^0	x_{n1}			

where each x_{ij} is an element in the i^{th} row and j^{th} column.

Note: after the first two rows, each successive row has one fewer calculated column than the previous row. Therefore, the array shrinks by one column each row we travel down the array, until the final row has only a single column.

For example

Example — Generic 1

Given: $A(s) = a_3 s^3 + a_2 s^2 + a_1 s + a_0$

Find the Routh Array

s^3	a_3	a_1
s^2	a_2	a_0
s^1	x_{31}	x_{32}
s^0	x_{41}	

Now for the calculated elements x_{ij} , construct a sub-matrix for each unknown element based on the elements in the two previous rows. Always use the elements in the first column and elements in the column to the right of the unknown element to be calculated.

For element x_{ij} the sub-matrix is

$$\begin{bmatrix} x_{(i-2)1} & x_{(i-2)(j+1)} \\ x_{(i-1)1} & x_{(i-1)(j+1)} \end{bmatrix}$$

If $i = 3$ and $j = 1$, the sub-matrix is

$$\begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix}$$

note, in this example, $x_{11} = a_n$, $x_{12} = a_{n-2}$, $x_{21} = a_{n-1}$, and $x_{22} = a_{n-3}$.

Finally, if $i = 7$ and $j = 4$, the sub-matrix is

$$\begin{bmatrix} x_{51} & x_{55} \\ x_{61} & x_{65} \end{bmatrix}$$

Once the sub-matrix is determined, the element to be calculated is simply the negative of the determinant divided by the element $x_{(i-1)1}$

$$x_{ij} = \frac{- \begin{vmatrix} x_{(i-2)1} & x_{(i-2)(j+1)} \\ x_{(i-1)1} & x_{(i-1)(j+1)} \end{vmatrix}}{x_{(i-1)1}}$$

where

$$\begin{vmatrix} x_{(i-2)1} & x_{(i-2)(j+1)} \\ x_{(i-1)1} & x_{(i-1)(j+1)} \end{vmatrix} = x_{(i-2)1} \cdot x_{(i-1)(j+1)} - x_{(i-2)(j+1)} \cdot x_{(i-1)1}$$

so

$$x_{ij} = \frac{x_{(i-2)(j+1)} \cdot x_{(i-1)1} - x_{(i-2)1} \cdot x_{(i-1)(j+1)}}{x_{(i-1)1}}$$

Once again, the math looks more complicated than it really is. Let's consider a few examples

Example — Generic 1, Reboot

For this example, we'll finish the previous one.

Given: $A(s) = a_3s^3 + a_2s^2 + a_1s + a_0$

Find: The stability of the system

The Routh Array

s^3	a_3	a_1
s^2	a_2	a_0
s^1	x_{31}	x_{32}
s^0	x_{41}	

where

$$x_{31} = \frac{- \begin{vmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{vmatrix}}{x_{21}} = \frac{- \begin{vmatrix} a_3 & a_1 \\ a_2 & a_0 \end{vmatrix}}{a_2} = \frac{a_2 \cdot a_1 - a_3 \cdot a_0}{a_2}$$

$$x_{32} = \frac{-\begin{vmatrix} x_{11} & x_{13} \\ x_{21} & x_{23} \end{vmatrix}}{x_{21}} = \frac{-\begin{vmatrix} a_3 & 0 \\ a_2 & 0 \end{vmatrix}}{a_2} = \frac{a_2 \cdot a_1 - a_3 \cdot a_0}{a_2} = 0$$

Note: Pad the array with zeros during the x_{ij} calculations.

$$x_{41} = \frac{-\begin{vmatrix} x_{21} & x_{22} \\ x_{31} & x_{32} \end{vmatrix}}{x_{31}} = \frac{-\begin{vmatrix} a_2 & a_0 \\ x_{31} & x_{32} \end{vmatrix}}{x_{31}} = \frac{x_{31} \cdot a_0 - a_2 \cdot x_{32}}{x_{31}} = \frac{x_{31} \cdot a_0 - a_2 \cdot 0}{x_{31}} = a_0$$

The final Routh array

s^3	a_3	a_1
s^2	a_2	a_0
s^1	x_{31}	0
s^0	a_0	

Now, for the stability analysis: inspect the elements in the first column. Starting with the first row s^n moving down, compare the signs (positive or negative). The number of sign changes in the first column is equal to the number of unstable poles.

Example — Second Order

Given: $A(s) = s^2 - 3s - 4$

Find: The stability of the system with the Routh Array

s^2	1	-4
s^1	-3	0
s^0	x_{31}	

Note: element x_{22} is not defined from the characteristic polynomial, but will be needed for later calculations so it's set to zero.

Only x_{31} needs to be calculated to determine stability

$$x_{31} = \frac{-\begin{vmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{vmatrix}}{x_{21}} = \frac{-\begin{vmatrix} 1 & -4 \\ -3 & 0 \end{vmatrix}}{-3} = \frac{12}{-3} = -4$$

The first column

$$\begin{bmatrix} +1 \\ -3 \\ -4 \end{bmatrix}$$

There is only one sign change in the first column; therefore, there is one unstable pole.

Example — Second Order

Given: $A(s) = s^2 + 5s + 4$

Find: The stability of the system with the Routh Array

s^2	1	4
s^1	5	0
s^0	x_{31}	

Only x_{31} needs to be calculated to determine stability

$$x_{31} = -\frac{\begin{vmatrix} 1 & 4 \\ 5 & 0 \end{vmatrix}}{5} = \frac{20}{5} = 4$$

So, the first column

$$\begin{bmatrix} +1 \\ +5 \\ +4 \end{bmatrix}$$

There are no sign changes in the first column; therefore, there are no unstable poles.

Example — Second Order

Given: $A(s) = s^2 - 5s + 4$

Find: the stability of the system with the Routh Array

s^2	1	4
s^1	-5	0
s^0	x_{31}	

Again, only x_{31} needs to be calculated to determine stability

$$x_{31} = -\frac{\begin{vmatrix} 1 & 4 \\ -5 & 0 \end{vmatrix}}{-5} = \frac{-20}{-5} = 4$$

So, the first column

$$\begin{bmatrix} +1 \\ -5 \\ +4 \end{bmatrix}$$

Now, there are two sign changes in the first column; therefore, there are two unstable poles. Since this is a second order system, all poles are unstable.

Example — Tunable Parameter

Given: $A(s) = s^3 + 9s^2 + Ks + 12$, where $-\infty < K < \infty$

Find: The stability of the system

Find: The range of K such that the system is stable

s^3	1	k
s^2	9	12
s^1	x_{31}	x_{32}
s^0	x_{41}	

Again, only x_{31} needs to be calculated to determine stability

$$x_{31} = \frac{-\begin{vmatrix} 1 & K \\ 9 & 12 \end{vmatrix}}{9} = \frac{9K - 12}{9}$$

$$x_{32} = \frac{-\begin{vmatrix} 1 & 0 \\ 9 & 0 \end{vmatrix}}{9} = 0$$

$$x_{41} = \frac{-\begin{vmatrix} 9 & 12 \\ x_{31} & 0 \end{vmatrix}}{x_{31}} = \frac{12x_{31}}{x_{31}} = 12$$

So, the first column

$$\begin{bmatrix} +1 \\ +9 \\ x_{31} \\ +12 \end{bmatrix}$$

For the system to be stable $x_{31} > 0$

$$x_{31} = \frac{9K - 12}{9} > 0$$

Therefore,

$$9K - 12 > 0$$

or

$$K > \frac{12}{9} = 1\frac{1}{3}$$

The range of stability is then

$$1\frac{1}{3} < K < \infty$$

Any value of K outside of this range produces an unstable system.

7.3.1 Advanced Analysis

There are four cases the Routh array analysis will fall into

1. Nominal

- This is the case for all previous Routh examples — basically, everything works as expected

2. Zero element in the first column

- Either from the characteristic polynomial or from a calculated element, some $x_{i1} = 0$, with another element in the row being non-zero

3. A row with all zeros

- There exists a row with all zeros for elements.

4. Indeterminate array

- There is an array that cannot be solved. The Routh array is **sufficient** to determine stability, but not **necessary**.

Nominal

The complete Routh array is formed and the first column may be analyzed directly.

Example — Nominal

Given: $A(s) = s^4 + 16s^3 + 70s^2 + 196s - 816$

Find: The system stability with the Routh Array

s^4	1	70	-816
s^3	16	196	0
s^2	x_{31}	x_{32}	x_{33}
s^1	x_{41}	x_{42}	
s^0	x_{51}		

$$x_{31} = \frac{- \begin{vmatrix} 1 & 70 \\ 16 & 196 \end{vmatrix}}{16} = \frac{924}{16} = 57.75$$

$$x_{32} = \frac{- \begin{vmatrix} 1 & -816 \\ 16 & 0 \end{vmatrix}}{16} = \frac{-13056}{16} = -816$$

$$x_{33} = \frac{- \begin{vmatrix} 1 & 0 \\ 16 & 0 \end{vmatrix}}{16} = \frac{0}{16} = 0$$

$$x_{41} = \frac{- \begin{vmatrix} 16 & 196 \\ 57.75 & -816 \end{vmatrix}}{57.75} = \frac{24375}{57.75} = 422.08$$

$$x_{42} = \frac{- \begin{vmatrix} 16 & 0 \\ 57.75 & 0 \end{vmatrix}}{57.75} = \frac{0}{57.75} = 0$$

$$x_{51} = \frac{- \begin{vmatrix} 57.75 & -816 \\ 422.08 & 0 \end{vmatrix}}{422.08} = \frac{-344420}{422.08} = -816$$

The final Routh Array

s^4	1	70	-816
s^3	16	196	0
s^2	57.75	-816	0
s^1	422.08	0	
s^0	-816		

There is a single sign change in the first column; therefore, the system has one unstable pole (among four poles) and is unstable.

Zero Element in the First Column

Initially, the Routh Array cannot be completed because a zero in the first column produces subsequent infinite elements.

Example — Zero Element

Given: $A(s) = s^5 + 2s^4 + 2s^3 + 4s^2 + 11s + 10$

Find: The system stability using the Routh Array

s^5	1	2	11
s^4	2	4	10
s^3	x_{31}	x_{32}	x_{33}
s^2	x_{41}	x_{42}	x_{43}
s^1	x_{51}	x_{52}	x_{53}
s^0	x_{61}	x_{62}	x_{63}

$$x_{31} = \frac{- \begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix}}{2} = \frac{0}{2} = 0$$

We have encountered a zero in the first row; however, we need to calculate more elements to determine if the entire row is zero or just the first element.

$$x_{32} = \frac{- \begin{vmatrix} 1 & 11 \\ 2 & 10 \end{vmatrix}}{2} = \frac{12}{2} = 6$$

Since $x_{32} \neq 0$ we have a zero element in the first column only case. The process is simple, set the zero element equal to a variable, ϵ , complete the array and then evaluate the elements in the first column as $\epsilon \rightarrow 0^+$. Update the Routh Array with the ϵ variable and the x_{32} element.

s^5	1	2	11
s^4	2	4	10
s^3	ϵ	6	x_{33}
s^2	x_{41}	x_{42}	x_{43}
s^1	x_{51}	x_{52}	x_{53}
s^0	x_{61}	x_{32}	x_{63}

Moving on

$$x_{33} = 0$$

by observation

$$x_{41} = \frac{- \begin{vmatrix} 2 & 4 \\ \epsilon & 6 \end{vmatrix}}{\epsilon} = \frac{4\epsilon - 12}{\epsilon}$$

$$x_{42} = \frac{- \begin{vmatrix} 2 & 10 \\ \epsilon & 0 \end{vmatrix}}{\epsilon} = \frac{10\epsilon}{\epsilon} = 10$$

$$x_{43} = \frac{- \begin{vmatrix} 2 & 0 \\ \epsilon & 0 \end{vmatrix}}{\epsilon} = 0$$

$$x_{51} = \frac{- \begin{vmatrix} \epsilon & 6 \\ x_{41} & 10 \end{vmatrix}}{x_{41}} = \frac{6x_{41} - 10\epsilon}{x_{41}}$$

$$x_{52} = \frac{- \begin{vmatrix} \epsilon & 0 \\ x_{41} & 0 \end{vmatrix}}{x_{41}} = 0$$

$$x_{61} = \frac{- \begin{vmatrix} x_{41} & 10 \\ x_{51} & 0 \end{vmatrix}}{x_{51}} = 10$$

Write the Routh Array once more with the calculated element that don't require limit analysis to determine their values

s^5	1	2	11
s^4	2	4	10
s^3	ϵ	6	0
s^2	x_{41}	10	0
s^1	x_{51}	0	
s^0	10		

The final evaluation

$$x_{31} = \lim_{\epsilon \rightarrow 0^+} \epsilon = +0$$

Since ϵ approaches zero from the positive side x_{31} is a small positive number.

$$x_{41} = \lim_{\epsilon \rightarrow 0^+} \frac{4\epsilon - 12}{\epsilon}$$

Let's divide numerator and denominator by ϵ and evaluate the limit

$$x_{41} = \lim_{\epsilon \rightarrow 0^+} \frac{4 - \frac{12}{\epsilon}}{1} = 4 - \frac{12}{0} \rightarrow -\infty$$

Then, finally

$$x_{51} \frac{6x_{41} - 10\epsilon}{x_{41}}$$

Divide numerator and denominator by x_{41} then evaluate

$$x_{51} \frac{6 - \frac{10\epsilon}{x_{41}}}{1} = 6 - \frac{10\epsilon}{x_{41}} = 6$$

the complete Routh Array

s^5	+1	2	11
s^4	+2	4	10
s^3	+0	6	0
s^2	$-\infty$	10	0
s^1	+6	0	
s^0	+10		

The first column

$$\begin{bmatrix} +1 \\ +2 \\ +0 \\ -\infty \\ +6 \\ +10 \end{bmatrix}$$

has two sign changes; therefore, the system is unstable with two poles (of five) in the unstable region.

7.3.2 Row of Zeros

This case occurs when there are symmetric poles about the origin in the S-plane. The approach is to replace the elements in the zero row with coefficients from an **auxiliary** equation. The auxiliary equation is derived from element in the row above the zero row.

Example — Row of Zeros

Given: $A(s) = s^4 + 12s^3 + 36s^2 + 48s + 128$

Find: The system stability with the Routh Array

s^4	1	36	128
s^3	12	48	0
s^2	32	128	0
s^1	0	0	
s^0			

Analysis stops after calculating row s^1 because all elements are zero. Look to the previous row for the coefficient of an auxiliary equation. Additionally, the row label will determine the order of the equation.

$$P(s) = 32s^2 + 128$$

Note: the coefficient for $P(s)$ (the auxiliary equation) are directly from the elements in the previous row, plus the highest order of $P(s)$ is the row label. The next element is blueuced by two.

Now, to determine the replacement element, take the firs derivative of $P(s)$

$$\frac{dP(s)}{ds} = 64s + 0$$

so, $x_{41} = 64$ and $x_{42} = 0$. We replace the zeros in the array and continue with completing the array.

s^4	1	36	128
s^3	12	48	0
s^2	32	128	0
s^1	64	0	
s^0			

which leaves on x_{51} to calculate

$$x_{51} = \frac{- \begin{vmatrix} 32 & 128 \\ 64 & 0 \end{vmatrix}}{64} = 128$$

The final Routh Array

s^4	1	36	128
s^3	12	48	0
s^2	32	128	0
s^1	64	0	
s^0	128		

with the first column

$$\begin{bmatrix} +1 \\ +12 \\ +32 \\ +64 \\ +128 \end{bmatrix}$$

Therefore, the system is stable with no sign changes in the first column. There is additional information we can learn from the auxiliary equation.

Solve for the roots of $P(s)$ to determine the symmetric poles.

$$P(s) = 32s^2 + 128 = 0$$

$$32s^2 = -128$$

$$s^2 = \pm \sqrt{-\frac{128}{32}} = \pm \sqrt{-4} = \pm 2j$$

Of the four poles in the charactering polynomial two are in the imaginary axis at $\pm 2j$. The remaining two poles are in the stable region.

7.3.3 Indeterminate Array

As previously stated, the Routh Array is sufficient to prove stability, but not necessary. If we encounter a characteristic polynomial that we are unable to produce a complete array, we cannot use the failure to the Routh analysis to make a claim as to the stability of the system.

7.3.4 Solutions with Multiple Approaches

Example — Special Case

Given: $A(s) = s^3 - 9s$

Find: The system stability with the Routh Array

For this problem, we have three options

Option 1 — Direct with Auxiliary Equation

Calculate the array of $A(s) = s^3 + 0s^2 - 9s + 0$

Determine the elements in the array as before

s^3	1	-9
s^2	0	0
s^1		
s^0		

At s^2 we have encountered a row of all zeros, so we to calculate the auxiliary equation and its derivative

$$P(s) = s^3 - 9s$$

and

$$\frac{dP(s)}{ds} = 3s^2 - 9$$

Now

s^3	1	-9
s^2	3	-9
s^1	x_{31}	0
s^0	x_{41}	

$$x_{31} = \frac{-\begin{vmatrix} 1 & -9 \\ 3 & -9 \end{vmatrix}}{3} = \frac{-27 + 9}{3} = -6$$

$$x_{41} = \frac{-\begin{vmatrix} 3 & -9 \\ -6 & 0 \end{vmatrix}}{-6} = -9$$

Then evaluation of the first column of the final array

s^3	1	-9
s^2	3	-9
s^1	-6	0
s^0	-9	

reveals one unstable pole in the system.

Option 2 — Direct with Modified Characteristic Polynomial & Limit Analysis

Given: $A(s) = s^3 - 9s$

Factor the common s from the characteristic polynomial such that

$$A(s) = s(s^2 - 9)$$

and apply the Routh array only to the expression within the parentheses

$$\hat{A}(s) = s^2 - 9 = s^2 + 0s - 9$$

s^2	1	-9
s^1	0	0
s^0		

in the s^1 row we have all zeros, so we find the auxiliary equation

$$P(s) = s^2 - 9$$

and its derivative

$$\frac{dP(s)}{ds} = 2s + 0$$

Replacing the zero row with the auxiliary equation coefficients produces a final array

s^2	1	-9
s^1	2	0
s^0	-9	

Evaluating the first column, we find the system $\hat{A}(s)$ is unstable. Since $A(s) = s(\hat{A}(s))$ it is unstable as well. Additionally, we know the original characteristic equation has one marginally stable pole at the origin. Our analysis tells us the original characteristic equation has one pole at the origin, and two pole symmetric about the origin. The symmetric pole can be found by solving $P(s) = 0$ for s . The solution is $s = \pm 3$.

There exists some special occurrences where either the limit analysis (zero in the first row) and the auxiliary equation analysis (a row with all zeros) can be used to evaluate the stability. Moreover, as seen above, we can perform the analysis on modified versions of the characteristic equation. The trick is to be consistent, follow the process, and remember what you've done.